ADVANCED SQL DATE Functions

Exercise 6

Table 1: Employees

```
Q1. SELECT  emp_id,
            name,
            hire_date,
            DATEADD(MONTH,6,hire_date) AS hire_plus_6_months
FROM        Employees;
```

emp_id	name	hire_date	hire_plus_6_months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Table 2: Students

```
Q2. SELECT  student_id,
            name,
            DATEDIFF(DAY, dob, '2026-04-21') AS age_in_days
FROM        Students;
```

student_id	name	age_in_days
101	Maya	7559
102	Ethan	7812
103	Sienna	7342

Table 3: Events

```
Q3. SELECT  event_id,
            event_name,
            DATEDIFF(DAY, CURRENT_DATE, event_date) AS days_remaining
FROM        Events;
```

event_id	event_name	days_remaining
1	Seminar	-675
2	Workshop	-232
3	Hackathon	-456

Table 4: Invoices

```
Q4. SELECT invoice_id,
           issue_date,
           due_date,
           DATEDIFF(DAY, issue_date, due_date) AS days_between
    FROM   Invoices;
```

invoice_id	issue_date	due_date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Table 5: Courses

```
Q5. SELECT  course_id,
            name,
            TO_CHAR(start_date, 'Month YYYY') AS formatted_date
    FROM    Courses;
```

course_id	name	formatted_date
201	SQL Basics	May 2025
202	Python	June 2025

Table 6: Memberships

```
Q6. SELECT member_id,
           DATE_FROM_PARTS(start_year, start_month, start_day) AS full_start_date
    FROM   Memberships;
```

member_id	full_start_date
1	2023-05-10
2	2022-11-25

Table 7: Subscriptions

Q7. SELECT sub_id,
plan,
DATEADD(YEAR, 1, renewal_date) AS extended_renewal_date
FROM Subscriptions;

sub_id	plan	extended_renewal_date
11	Basic	2026-01-01
12	Premium	2026-03-15

Table 8: Orders

Q8. SELECT order_id,
order_date,
~~today_date~~, CURRENT_DATE AS today_date,
DATEDIFF(DAY, order_date, CURRENT_DATE) AS days_since_order
FROM Orders;

order_id	order_date	today_date	days_since_order
1001	2025-04-15	2026-04-21	371
1002	2025-04-10	2026-04-21	376

Table 9: Trainings

Q9. SELECT training_id,
topic,
EXTRACT(YEAR FROM training_date) AS training_year
FROM Trainings;

training_id	topic	training_year
1	Safety	2025
2	Compliance	2025

Table 10: Blog_Posts

```
Q10. SELECT post_id,
             title,
             EXTRACT(hour FROM published_on) AS hour_published,
             EXTRACT(minute FROM published_on) AS minute_published,
FROM  Blog_Posts;
```

post_id	title	hour_published	minute_published
1	SQL Tips	10	15
2	Data Cleaning	16	45

Table 11: Drivers

```
Q11. SELECT driver_id,
             license_expiry,
             DATEDIFF(DAY, CURRENT_DATE, license_expiry) AS days_left
     FROM    Drivers;
```

driver_id	license_expiry	days_left
301	2025-08-10	-254
302	2023-12-31	-842

Table 12: Messages

```
Q12. SELECT message_id,
             sent_timestamp,
             CURRENT_TIMESTAMP AS
             current_timestamp,
             DATEDIFF('second', sent_timestamp, CURRENT_TIMESTAMP) AS
             seconds_since_sent
     FROM    Messages;
```

message_id	sent_timestamp	current_timestamp	seconds_since_sent
1	2025-04-19 09:32:45	2026-04-21 15:00:00	31,728,435
2	2025-04-18 23:59:59	2026-04-21 15:00:00	31,762,801

Table 13: Returns

```
Q13. SELECT return_id,
             return_date
             DATEADD(day, 15, return_date) AS restock_date
     FROM    Returns;
```

return_id	return_date	restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Table 14: Assignments

```
Q14. SELECT assign_id,
             TO_DATE(assigned_on) AS assigned_on_date
     FROM    Assignments;
```

assign_id	assigned_on_date
1	2025-03-01
2	2025-03-05

Table 15: Meetings

```
Q15. SELECT meeting_id,
             TO_CHAR(scheduled_time, 'Month DD, YYYY at HH12:MI AM')
             AS formatted_meeting_time
     FROM    Meetings;
```

meeting_id	formatted_meeting_time
1	April 19, 2025 at 02:00 PM
2	April 19, 2025 at 09:30 AM